```
python -m pip --version
python -m pip install selenium
```

P6

Install Selenium WebDriver and Configure Environment Using Python

```
from selenium import webdriver

driver = webdriver.Chrome()

driver.get("https://www.google.com")

print("Page Title:", driver.title)

driver.quit()
```

P7

Write a Selenium WebDriver Script to Automate Google Search

```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.common.keys import Keys

driver = webdriver.Chrome()
driver.get("https://www.google.com")

search_box = driver.find_element(By.NAME, "q")
search_box.send_keys("Selenium WebDriver")
search_box.send_keys(Keys.ENTER)
print("Page Title:", driver.title)
print("Current URL:", driver.current_url)

driver.quit()
```

P8

Automate Form Handling Using Selenium (Login Page Automation)

```
from selenium import webdriver
from selenium.webdriver.common.by import By
import time

driver = webdriver.Chrome()
driver.get("https://the-internet.herokuapp.com/login")
time.sleep(3)

username = driver.find_element(By.ID, "username")
username.send_keys("tomsmith")
time.sleep(2)

password = driver.find_element(By.ID, "password")
password.send_keys("SuperSecretPassword!")
time.sleep(2)

login_button = driver.find_element(
   By.CSS_SELECTOR, "button[type='submit']"
)
login_button.click()
time.sleep(5)
print("Page Title:", driver.title)
print("Current URL:", driver.current_url)

message = driver.find_element(By.ID, "flash").text
print("Login Message:", message)

time.sleep(3)
```

```
driver.quit()
```

P9

Implement Selenium Scripts Using Different Locators

```
from selenium import webdriver
from selenium.webdriver.common.by import By
import time
driver = webdriver.Chrome()
driver.get("https://the-internet.herokuapp.com/login")
time.sleep(3)

# ID Locator
username = driver.find_element(By.ID, "username")
username.send_keys("tomsmith")
time.sleep(2)

password = driver.find_element(By.ID, "password")
password.send_keys("SuperSecretPassword!")
time.sleep(2)

# XPath Locator
login_button = driver.find_element(
   By.XPATH, "//button[@type='submit']")

login_button.click()
time.sleep(5)
print("Page Title:", driver.title)
print("Current URL:", driver.current_url)

# CSS Selector
driver.get("https://the-internet.herokuapp.com/login")
```

```
time.sleep(3)

username_css = driver.find_element(
   By.CSS_SELECTOR, "#username")
username_css.send_keys("tomsmith")

password_css = driver.find_element(
   By.CSS_SELECTOR, "#password")
password_css.send_keys("SuperSecretPassword!")

login_css = driver.find_element(
   By.CSS_SELECTOR, "button[type='submit']")

login_css.click()
time.sleep(5)
print("CSS Selector Test Completed")

# Name Locator demonstration
driver.get("https://www.google.com")
time.sleep(3)

search_box = driver.find_element(By.NAME, "q")
search_box.send_keys("Selenium WebDriver")
time.sleep(3)

from selenium.webdriver.common.keys import Keys
search_box.send_keys(Keys.ENTER)
time.sleep(5)
print("Name Locator Test Completed")
print("Page Title:", driver.title)
```

```
driver.quit()
```

P10

Handle Alerts, Frames and Multiple Windows Using Selenium

A. HANDLING ALERTS

```
from selenium import webdriver
from selenium.webdriver.common.by import By
import time

driver = webdriver.Chrome()

driver.get("https://the-internet.herokuapp.com/javascript_alerts")
time.sleep(3)

driver.find_element(
    By.XPATH,
    "//button[text()='Click for JS Alert']"
).click()

time.sleep(2)

alert = driver.switch_to.alert

print("Alert Text:", alert.text)

alert.accept()

time.sleep(3)

driver.quit()
```

B. HANDLING FRAMES

```
from selenium import webdriver

from selenium.webdriver.common.by import By

from selenium.webdriver.common.keys import Keys

from selenium.webdriver.common.action_chains import ActionChains

import time


driver = webdriver.Chrome()

driver.get("https://the-internet.herokuapp.com/iframe")

time.sleep(3)



driver.switch_to.frame("mce_0_ifr")

time.sleep(2)


body = driver.find_element(By.TAG_NAME, "body")


body.click()

time.sleep(1)


actions = ActionChains(driver)

actions.key_down(Keys.CONTROL).send_keys("a").key_up(Keys.CONTROL)

actions.send_keys(Keys.BACKSPACE)

actions.send_keys("Hello Selenium")

actions.perform()

time.sleep(5)


driver.switch_to.default_content()

time.sleep(2)

driver.quit()
```

C. HANDLING MULTIPLE WINDOWS

```
from selenium import webdriver
from selenium.webdriver.common.by import By
import time

driver = webdriver.Chrome()

driver.get("https://the-internet.herokuapp.com/windows")
time.sleep(3)

driver.find_element(
    By.LINK_TEXT, "Click Here"
).click()

time.sleep(3)

windows = driver.window_handles

print("Number of Windows:", len(windows))

driver.switch_to.window(windows[1])

time.sleep(5)

print("New Window Title:", driver.title)
print("New Window URL:", driver.current_url)

time.sleep(3)
```

```
driver.quit()
```